

Multi-View Basketball Player and Ball Tracking with 3D Reconstruction

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Abstract. We present a multi-camera pipeline for tracking players, referees and the ball during a basketball action recorded at the Sanbapolis facility, and for reconstructing their 3D trajectories via triangulation. Players and referees are detected and tracked independently on each camera view, at the native frame rate, with a general-purpose detector and its built-in tracker. The ball is a much harder target: its small size and frequent blur or occlusion defeat a generic detector almost entirely, and even fine-tuning it for the ball class alone, combined with tiled inference, only partially closes the gap. We instead compare a classical motion-and-colour baseline against a pretrained, basketball-specific detector, the latter clearly outperforming the former. Camera intrinsics and extrinsics, provided for this project, undistort each view and triangulate cross-camera-matched 2D detections into a court-centered 3D reconstruction, with correspondence resolved geometrically through a reprojection-error voting scheme rather than appearance. Both the 2D detections and the 3D reconstruction are evaluated against manually annotated ground truth, following standard detection/tracking protocols in 2D and a distance-based matching protocol in 3D that also captures trajectory-level continuity.

1 Introduction

This work addresses the Player Tracking project of the Computer Vision course: given a multi-camera setup at the Sanbapolis facility, detect and track players, referees and the ball in each view, and reconstruct their 3D trajectories by triangulation. One action was recorded from three fixed, synchronized cameras (`out2`, `out4`, `out13`) on a FIBA-regulation 28×15 m court. As a team of two, the mandatory scope is annotation (task 1), 2D tracking with evaluation (task 2/2a), and 3D triangulation with evaluation (task 3/3a) using at least two views; we use all three throughout.

Two aspects make this work not straightforward: the ball is small, fast and often blurred or occluded, which turns plain detection into most of the effort (Sec. 2.3); and cross-camera association cannot reuse a per-camera track ID, so correspondence has to come from projective geometry instead, since no appearance-based re-identification was used (future work, Sec. 4). The rest of this report covers the dataset, the camera model, 2D detection/tracking, and cross-camera 3D triangulation, each evaluated quantitatively – including a trajectory-level continuity measure – in Sec. 3.

2 Method

2.1 Dataset and Annotation

The action used for this project was annotated on Roboflow at 5 fps across the three synchronized views: the ball, the two referees, and every player individually, identified by team colour and jersey number rather than pooled under one generic label – which later makes it possible to match a specific person across cameras for the 3D pseudo-ground-truth (Sec. 3.3), while all non-ball categories are pooled into a single **player** class for the 2D evaluation (Sec. 3.2). The annotation effort was split across the groups taking the course, each assigned a different action from the same three-camera setup and contributing its boxes to a shared dataset; besides the images used here for evaluation, that pooled, multi-action dataset was later reused to fine-tune a ball-specific detector, discussed next.

2.2 Camera Model

Per-camera intrinsics (K , distortion) and extrinsics (rotation/ translation w.r.t. a court-centered frame) were provided for this project rather than self-calibrated. They undistort every frame before 2D detection (Sec. 2.3) and build each camera’s projection matrix $P = K[R | t]$ for triangulation (Sec. 2.4); the full undistortion+triangulation chain is validated end-to-end in Sec. 3.3 instead of a separate reprojection-error figure.

2.3 2D Detection and Tracking

Players and referees. Each camera is processed independently at the native frame rate: frames are undistorted using the calibration from Sec. 2.2 (a per-pixel remap built once and reused for every frame), resized, and passed to a pretrained YOLO detector [6], restricted to the COCO **person/ sports ball** classes, run through Ultralytics’ built-in tracking mode, which defaults to BoT-SORT [3] when no alternative tracker is configured, as here. A low confidence threshold favors recall, relying on the tracker’s temporal persistence to filter out spurious detections over time.

Ball: from generic detector to fine-tuning. The same player-tracking pipeline is effectively blind to the ball: after the required resize it covers only a handful of pixels, comparable to or smaller than the backbone’s convolutional stride, so most of its signal is gone before the detection head sees it. The first fix fine-tuned YOLO on the ball class alone, using the pooled, multi-action dataset from Sec. 2.1, plus SAHI tiled inference [2] for higher effective resolution per tile – which helped only marginally, the ball still collapsing to a few pixels within a tile on the wider-shot cameras. This confirmed a structural bottleneck rather than a training one, so the route was abandoned for the two alternatives below.

Ball: classical baseline. The first alternative drops the detector for a classical, motion-based pipeline: MOG2 background subtraction (with shadow detection) isolates moving foreground, shadow pixels are thresholded away from its soft output, and the mask is cleaned with a closing (reconnecting the ball’s blob) then an opening (removing noise). Surviving contours are filtered by area (per-camera, tuned to the ball’s expected size), shape (circularity, near-square aspect ratio), and colour (a minimum fraction of pixels in the basketball’s HSV hue range); whatever passes all three is that frame’s detection, with no tracker on top. It needs no training data or GPU, but inherits background subtraction’s weaknesses: it confuses other fast, similarly shaped or coloured objects for the ball, and misses it while briefly static or occluded.

Ball: pretrained specialized detector. The best-performing alternative is WASB/HRNet [4], a heatmap-based detector on an HRNet backbone, used zero-shot with its basketball-specific checkpoint rather than retrained here. It stacks a short window of consecutive frames channel-wise, and decodes the output heatmap via connected-component analysis, so a detection is a blob’s weighted centroid rather than a raw peak. Its default threshold, calibrated for broadcast footage, rejects almost everything on our recordings; after manually confirming the residual low-confidence peaks did correspond to the ball, we lowered it to a usable value. On the widest-shot camera the same tiling idea works this time: overlapping tiles processed independently recover detections a full-frame pass misses, at the cost of slower inference.

2.4 Cross-Camera Association and 3D Triangulation

Per-camera tracks are undistorted using the calibration from Sec. 2.2. Cross-camera identity is resolved geometrically rather than by appearance: for every camera pair and frame, each candidate pairing of per-camera track IDs is scored by triangulating the two detections and measuring the combined reprojection error back into both views [1], and a Hungarian assignment picks the best pairing per frame among pairs below an error threshold. A global identity only forms from pairs that are *mutually* the best match and win this assignment in several distinct frames – a single lucky frame is not enough evidence – and consistent pairs are then merged into cross-camera groups with a union-find structure, discarding any merge that would place two different IDs from the same camera in one group.

For every frame, detections sharing a global ID from at least two cameras are triangulated with a multi-view DLT solve. A court-sized plausibility box discards triangulated points that fall outside it, which removes residual wrong matches rather than genuine players.

3 Results

3.1 Experimental Setup

The 2D evaluation matches predicted/ground-truth boxes per frame (per camera, per class) with a Hungarian assignment on IoU, reporting Precision, Recall,

F1 and MOTP (mean IoU over matched pairs), following CLEAR MOT [5]. The acceptance threshold is deliberately low, so Precision/Recall/F1 mainly reflect whether an object was found in roughly the right place, while MOTP carries the localization-quality signal. The 3D evaluation triangulates annotated boxes (≥ 2 views, same frame, same category) into a pseudo-ground-truth 3D point, matched to the reconstruction with a Hungarian assignment on 3D distance (threshold 3 m).

3.2 2D Tracking

Table 1 reports player/referee accuracy per camera and the ball trackers pooled over all three (per-camera ball numbers follow the same `out13`-is-hardest pattern). `out2/out4` (closer, narrower shots) track players considerably better than `out13` (wide, more/smaller/more-occluded people); pooled, Recall = 0.712, Precision = 0.775, MOTP = 0.423. For the ball, WASB roughly doubles F1 over the classical baseline with fewer false positives, but both miss most instances – even WASB finds fewer than 1 in 6 – consistent with the small-object argument from Sec. 2.3.

Table 1: 2D tracking accuracy: players+referees pooled (class_id 0) per camera and aggregated; ball trackers (class_id 1) pooled over all three cameras.

		TP	FP	FN	Precision / Recall / F1	MOTP
Players	<code>out2</code>	583	44	179	0.930 / 0.765 / 0.839	0.441
	<code>out4</code>	459	98	172	0.824 / 0.727 / 0.773	0.470
	<code>out13</code>	848	408	414	0.675 / 0.672 / 0.674	0.386
	All	1890	550	765	0.775 / 0.712 / 0.742	0.423
Ball	Classic (MOG2+HSV)	16	263	175	0.057 / 0.084 / 0.068	0.116
	WASB/HRNet	25	133	166	0.158 / 0.131 / 0.143	0.083

3.3 3D Reconstruction

Of 1085 pseudo-ground-truth 3D points, 487 (44.9% coverage) match a reconstructed point within 3 m, with an overall MED of 0.722 m and RMSE of 1.013 m. Per-identity (Table 2), most players/referees localize with sub-meter accuracy, with two exceptions: the ball (only 3 matches, limited by the same detection scarcity as Sec. 3.2) and `Red_23` (1.91 m over 75 matches – not a handful of outliers, pointing to a systematic cross-camera association problem for that identity).

3.4 Trajectory Continuity

Task 3a also calls for trajectory-level continuity, not just point accuracy. For each of the 18 global 3D tracks, *continuity* is the fraction of frames within its own span with a reconstructed position (1.0 = no gaps): 88.1% on average. More tellingly, 18 IDs exceeds the ~ 12 –14 people on court: vote-based re-identification (Sec. 2.4) sometimes fragments one person into several tracks once cross-camera correspondence is lost, and never re-merges them – a limitation of an association scheme with no appearance model.

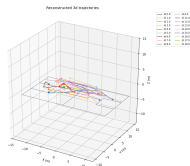


Fig. 1: Reconstructed 3D trajectories, court-centered frame; each colour is one global cross-camera identity (Sec. 2.4).

Table 2: 3D error by identity, sorted by mean error (two columns for space).

Identity	Mean [m]	n	Identity	Mean [m]	n
Red_2	0.307	29	White_14	0.592	26
Refree_1	0.308	49	White_2	0.614	32
Red_9	0.381	14	Red_7	0.668	11
White_27	0.460	45	White_34	0.678	71
White_16	0.476	82	Ball	1.431	3
Red_11	0.496	50	Red_23	1.907	75

4 Conclusions and Future Work

We built and evaluated a three-camera pipeline for player/referee and ball tracking with 3D reconstruction, covering the mandatory tasks for a two-person team using all three views rather than the two required. Players/referees reach moderate-to-good 2D accuracy and sub-meter 3D median error; the ball and Red_23 are the two clear outliers. The ball proved hard throughout: fine-tuning plus tiling was abandoned once diagnosed as structurally limited by object size, and even the better pretrained model (WASB) stays well short of usable. Future work includes diagnosing the Red_23 outlier, adding appearance-based re-identification against the fragmentation above, adopting identity-aware MOT metrics, and, as a bonus, importing the motion into Unreal Engine (task 4).

References

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